

L298N Pin	ESP32 Pin	Purpose
ENA (Enable A)	GPIO 13	Left Motor Speed (PWM)
IN1	GPIO 14	Left Motor Forward
IN2	GPIO 27	Left Motor Backward
IN3	GPIO 26	Right Motor Forward
IN4	GPIO 25	Right Motor Backward
ENB (Enable B)	GPIO 12	Right Motor Speed (PWM)
Power Module Terminal	L298N Terminal	Purpose
12V Output	12V (VCC)	Motor-ku thevayaana main power.
GND (Ground)	GND	Circuit-ah complete panna.
Ultrasonic Pin	ESP32 Pin	Purpose
Trig	GPIO 5	ESP32 signal-ah "Send" panna.
Echo	GPIO 18	ESP32 thirumba signal-ah "Receive" panna.
Ultrasonic Pin	Power Module / Board Pin	Note
VCC	5V Output (Power Module)	Sensor-ku steady 5V venum.
GND	GND (Power Module)	Common ground line-la connect pannunga.
LCD (I2C) Pin	Connection Point	Purpose
GND	Power Module GND	Common Ground
VCC	Power Module 5V	LCD backlight and display power
SDA	ESP32 GPIO 21	I2C Data line
SCL	ESP32 GPIO 22	I2C Clock line
Buzzer 1 Pin	Connection Point	Note
Positive (+ / Long Leg)	GPIO 19	ESP32-oda indha pin dhaan buzzer-ah ON/OFF pannum.
Negative (- / Short Leg)	GND	ESP32-oda GND pin-la connect pannunga.

Buzzer 2	Connection Point	Note
Buzzer 2 Positive (+)	Relay NO (Normally Open)	Switch ON aana dhaan buzzer adikkum.
Buzzer 2 Negative (-)	Power Module GND	Direct-ah ground-la kudunga.
Power Module 5V	Relay COM (Common)	Buzzer-ku thevayaana current inga dhaan pogum.
Relay Pin	Connection Point (ESP32 Dev B	Purpose
VCC	Power Module 5V	Relay coil power (Steady 5V).
GND	GND (ESP32 or Power Module)	Common Ground.
IN1	GPIO 4 (ESP32 Dev Board)	Buzzer 2-va activate panna indha signal pin.
IN2	GPIO 2 (ESP32 Dev Board)	(Optional) Future-la spray pump fix panna use aagum.
ESP32 CAM Pin	Power Module Pin	Note
5V	5V Output	Separate wire direct-ah module-la irundhu varanum.
GND	GND	Common ground line-la join pannunga.
ESP32 CAM Pin	ESP32 Dev Board Pin	Purpose
UOT (TX)	RX2 (GPIO 17)	Disease thagaval-ah "Brain" board-ku anuppa.
UOR (RX)	TX2 (GPIO 16)	(Optional) Brain kitta irundhu command vaanga.
L298N Terminal	Motor Side	Purpose
OUT 1	Left Motor (Wire 1)	Left motor-ah munnadi/pinnadi suthavaikka.
OUT 2	Left Motor (Wire 2)	
OUT 3	Right Motor (Wire 1)	Right motor-ah munnadi/pinnadi suthavaikka.
OUT 4	Right Motor (Wire 2)	